Heather Zinn Brooks

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Academic Positions

2020 – · · · ·
Harvey Mudd College
Assistant Professor
Department of Mathematics

2018 – 2020
University of California, Los Angeles
CAM Assistant Professor (postdoctoral position)
Department of Mathematics
Mentor: Mason Porter

Education

2018
Ph.D. Mathematics, University of Utah
Advisor: Paul Bressloff
Thesis title: *Dynamics and structure: from microtubule networks to population networks*

2012
B.S. Applied Mathematics, University of Utah
Advisor: Alla Borisyuk
Undergraduate Research Scholar
College of Science Research Award Finalist

Research Publications

Journal Articles

1. Hickok, A., Kureh, Y., **Brooks**, **H. Z.**, Feng, M., & Porter, M. A. (2021). A bounded-confidence model of opinion dynamics on hypergraphs. *SIAM Journal on Applied Dynamical Systems, in press*.
2. **Brooks**, **H. Z.**, & Porter, M. A. (2020). A model for the influence of media on the ideology of content in online social networks. *Physical Review Research*, *2*(2), 023041.
3. Carroll, S. R., **Brooks**, **H. Z.**, & Bressloff, P. C. (2020). Bifurcation analysis of pattern formation in a two-dimensional hybrid reaction–transport model. *Physica D: Nonlinear Phenomena*, *402*, 132274.
4. Wilson, S. N., Sindi, S. S., **Brooks**, **H. Z.**, Hohn, M. E., Price, C. R., Radunskaya, A. E., … Fefferman, N. H. (2020). How emergent social patterns in allogrooming combat parasitic infections. *Frontiers in Ecology and Evolution*, *8*, 54.
5. **Brooks**, **H. Z.**, & Bressloff, P. C. (2017). Turing mechanism for homeostatic control of synaptic density during *C. elegans* growth. *Physical Review E*, *96*(1), 012413.
6. **Brooks**, **H. Z.**, & Bressloff, P. C. (2016). A mechanism for Turing pattern formation with active and passive transport. *SIAM Journal on Applied Dynamical Systems*, *15*(4), 1823–1843.
7. **Brooks**, **H. Z.**, & Bressloff, P. C. (2015). Quasicycles in the stochastic hybrid Morris–Lecar neural model. *Physical Review E*, *92*(1), 012704.
8. Lawley, S. D., Tuft, M., & **Brooks**, **H. Z.** (2015). Coarse-graining intermittent intracellular transport: Two-and three-dimensional models. *Physical Review E*, *92*(4), 042709.

Books and Chapters

1. **Brooks, H. Z.**, Hohn, M. E., Price, C. R., Radunskaya, A. E., Sindi, S. S., Williams, N. D., … Fefferman, N. H. (2018). Mathematical analysis of the impact of social structure on ectoparasite load in allogrooming populations. In *Understanding complex biological systems with mathematics* (pp. 47–61). Springer.
2. Williams, N. D., **Brooks, H. Z.**, Hohn, M. E., Price, C. R., Radunskaya, A. E., Sindi, S. S., … Fefferman, N. H. (2018). How disease risks can impact the evolution of social behaviors and emergent population organization. In *Understanding complex biological systems with mathematics* (pp. 31–46). Springer.

Expository Articles & Outreach

1. **Brooks, H. Z.**, Chen, Y., Feng, M., Kureh, Y. H., Porter, M. A., & Volkening, A. (2020). How to move a SIAM minisymposium online from the comfort of your home. DSWeb: The Dynamical Systems Web.
2. **Brooks, H. Z.**, Kanjanasaratool, U., Kureh, Y. H., & Porter, M. A. (2020). Disease detectives: Using mathematics to forecast the spread of infectious diseases. Frontiers for Young Minds, Vol. 8: 577741.

Awards and Fellowships

2021–2024	**NSF DMS-2109239** *Advances in bounded-confidence models on networks* Principal Investigator, $208,684
2021	**OCAC Data Science Course Development Grant Award**, $1,500
2019–2021	**MAA Project NExT Fellowship**, $5,000
2017	**Stockham Medal for Conspicuously Effective Teaching nominee**, $1,000
	T. Benny and Gail T. Rushing Fellowship Award, $5,000
2016	**Center for Teaching and Learning Excellence Graduate Fellowship**, $15,000
2014–2016	**NSF Research Training Grant Fellowship RTG-1148230**, $20,457 per year
2014	**RTG Lab Rotation**, Neuronal Dynamics Laboratory (PI: John White), $5,000
Travel grants	**SIAM Early Career** (2019), $650
	AMS MRC (2018), $850
	SIAM Early Career (2018), $650
	Joint Mathematics Meeting (2017), $500
	SIAM Student (2016), $600
	NSF RTG (2015), $1000
	Organization for Computational Neurosciences (2013), $800

Teaching and Mentorship

Undergraduate Student Research

2021–· · · ·

- **Madelyn Andersen (HMC '22)**, *2021 Giovanni Borrelli Fellowship Recipient*. Graphon varieties.
- **Christina Catlett (Scripps '22)**, Multilayer bounded-confidence models with synchronous and asynchronous updating.
- **Andy Liu (HMC '23)**, Heterogeneous topic interdependencies in opinion dynamics.
- **Phousawanh Peaungvongpakdy (HMC '22)**, Media impact on opinion dynamics in adaptive social networks.

Teaching and Mentorship (continued)

- **Solomon Valore-Caplan (HMC '22)**, The effects of homophily and community structure on consensus in bounded-confidence models.
- **Ruixiao Wang (UCLA '22)**, Dynamics of the glass ceiling on networks (co-mentor Mason Porter).

2020–2021
- **Georgia Pope (HMC '21)**, Stability analysis of a mathematical model of hormonal contraception.
  Placement: MS Epidemiology, University of Washington
- **Sneha Sambandam (UCLA '21) and Annie Zhang (UCLA '21)**, Spread of competing content on networks with bounded-confidence dynamics.
  Placement: PhD CS, University of British Columbia (Sambandam) and MS Applied Math, University of Vermont (Zhang)

2019–2020
- **Gabrielle Lorenzi (UCLA '21)**, Networks, bias, and self-segregation of women and minorities in academia (co-mentor Mason Porter).
  Placement: Data Scientist, T. Rowe Price

2020
- **Siqiao Mu (Caltech '21)**, *Summer Undergraduate Research Fellowship program at Caltech*. SEIR models on networks (co-mentors Mason Porter and Franca Hoffmann).
  Placement: PhD ESAM, Northwestern University
- **Sujai Hiremath (Caltech '23)**, *Summer Undergraduate Research Fellowship program at Caltech*. Opinion formation on networks (co-mentors Mason Porter and Franca Hoffmann).

2019
- **Zehan Chao (UCLA graduate student), Zheyuan Cui (UCLA '21), Avery Edson (UCLA '20), Cesar Guajardo (Santa Monica College '20), Yihuan Huang (UCLA '21), Xingjia Wang (UCLA '20), Zhanyuan Yin (UCLA '21)**, *California Research Training Program in Computational and Applied Math*. Evaluating effectiveness of the Gang Reduction and Youth Development Program with dynamic mode decomposition and machine learning techniques.
- **Alex Pan (Caltech '22)**, *Summer Undergraduate Research Fellowship program at Caltech*. Opinion formation on networks (co-mentors Mason Porter and Franca Hoffmann).

2015
- **Marie Tuft (Utah '15)**, Quantitative analysis of virus trafficking in a biological cell (co-mentor Sean Lawley).
- **Oliver Richardson (Utah '17)**, Modeling learning on neural networks (co-mentor Sean Lawley).
- **Braden Schaer (Utah '15) and Anand Singh (Utah '16)**, Modeling diffusion of neurotransmitters (co-mentor Sean Lawley).

Clinic Program

2021–2022
- **Faculty Advisor, Policing Fairness Mathematics Clinic**.

2020–2021
- **Faculty Advisor, Estée Lauder CS/Mathematics Clinic**.

Courses

HMC
- **Math 189AB: Mathematics of Democracy**, Fall 2021
- **Math 73: Linear Algebra**, Spring 2021 (2 sections, online) and Spring 2022 (2 sections)
- **Math 131: Mathematical Analysis**, Fall 2020 (online)

UCLA
- **Math 168: Introduction to Networks**, Spring 2020
- **Math 142: Mathematical Modeling**, Fall 2019
- **Math 197: Reading Course in Nonlinear Dynamics**, Spring 2019

Teaching and Mentorship (continued)

- **Math 134: Linear and Nonlinear Systems of Differential Equations**, Fall 2018 and Winter 2019

U. Utah

- **Math 3140: Vector Calculus and Partial Differential Equations**, Spring 2018
- **Math 1210: Calculus 1**, Fall 2017 (online)
- **Math 2270: Linear Algebra**, Spring 2017
- **Math 3150: Partial Differential Equations for Engineers**, Summer 2016 and Fall 2016
- **Math 1050: College Algebra**, Summer 2013
- **Math 1010: Intermediate Algebra**, Spring 2013 and Spring 2014
- **Math 1030: Intro to Quantitative Reasoning**, Fall 2012

Organization of Conferences, Conference Sessions, and Workshops

September 2022

- **Building Networks: Women in Complex & Nonlinear Systems**, Workshop at Banff International Research Station (co-organizers Nina Fefferman, Nancy Rodríguez, Alexandria Volkening).

Summer 2022 and 2023

- **Data Science and Social Justice: Networks, Policy, and Education**, Summer Program at ICERM (organizing committee). Lead organizer for the associated ICERM workshop on Interdisciplinary Network Analysis Methods for Analyzing Social Systems.

April 2022

- **Southern California Applied Math Symposium**, Harvey Mudd College. Co-lead local organizer along with Jamie Haddock.

January 2022

- **Mathematics of Complex Systems**, SIAM special session at Joint Mathematics Meeting in Seattle, Washington (co-organizers Alexander Hoover, Alice Schwarze, Mason Porter, Alexandria Volkening).
- **Women in Math Biology**, AMS special session at Joint Mathematics Meeting in Seattle, Washington (co-organizers Christina Edholm, Maryann Hohn, Amanda Laubmeier, and Carrie Manore).

May 2021

- **Novel Techniques for Modeling Biological Systems in the Age of Data**, minisymposium at virtual SIAM conference on Applied Dynamical Systems (co-organizer Christopher Miles).

January 2021

- **Mathematical and Computational Methods for Complex Social Systems**, AMS Short Course at virtual Joint Mathematics Meeting (co-organizers Michelle Feng, Mason Porter, Alexandria Volkening).

May 2020

- **Topological Data Analysis and Data-Driven Modeling in Complex Systems**, minisymposium at virtual SIAM conference on the Mathematics of Data Science (co-organizer Mason Porter).

January 2020

- **Women in Mathematical Biology**, AMS-AWM special session at Joint Mathematics Meeting in Denver, Colorado (co-organizers Christina Edholm, Katharine Gurski, and Amanda Laubmeier).
- **Interactive Session on Fostering Equitable Classrooms**, Project NExT session at Joint Mathematics Meeting in Denver, Colorado (co-organizers Christina Edholm, Ryan Moruzzi, Mark Weaver).

November 2019

- **Disease Detectives: Stopping the Zombie Apocalypse with Mathematics**, booth at the UCLA Exploring Your Universe science festival on the role of mathematics and networks in epidemiology. Targeted toward families and children in the Los Angeles community (co-organizer Yacoub Kureh).

Organization of Conferences, Conference Sessions, and Workshops (continued)

May 2019

- **Dynamics of Democracy**, minisymposium at SIAM conference on Applied Dynamical Systems in Snowbird, Utah (co-organizer Alexandria Volkening).
- **Student and Postdoc Icebreaker Session**, introductory session for creating an inclusive environment for young researchers at SIAM conference on Applied Dynamical Systems in Snowbird, Utah (co-organizer Alexandria Volkening).

Invited Talks and Participation

Seminars and Colloquia

2021
- Little School Dynamics Colloquium, virtual
- Mathematical Sciences Colloquium, Appalachian State University, virtual
- Mathematics and Statistics Colloquium, College of the Holy Cross, virtual
- ESAM Colloquium, Northwestern University, virtual
- Mathematics Colloquium, Claremont Colleges, virtual

2020
- Dynamics Seminar, CU Boulder, virtual
- Women in Network Science Seminar, virtual
- Joint Applied Math Seminar, Claremont Colleges and University of Utah, virtual
- Networks Seminar, University of Houston, Houston, TX

2019
- Probability Seminar, Tulane University, New Orleans, LA
- Computational Math + X Seminar, Caltech, Pasadena, CA

2018
- Applied Math Seminar, Claremont Colleges, Claremont, CA
- Applied PDEs Seminar, Brigham Young University, Provo, UT

2017
- Biophysics Seminar, UCLA, Los Angeles, CA
- Howard Hughes Medical Institute Summer Research Program, Claremont, CA

Conference talks

2021
- Satellite on Higher-order Models in Network Science
  NetSci, virtual
- Minisymposium on Collective Behavior and Social Evolution
  Society for Mathematical Biology, virtual
- Minisymposium on Modeling Collective Behavior in Human Social Systems
  SIAM Conference on Applied Dynamical Systems, virtual
- Southern California Systems Biology Symposium, virtual
- Special Session on Agent-based Dynamics and Self-Organization in Biology
  Joint Mathematics Meeting, virtual
- Tutorial on Networks in Social Systems
  AMS Short Course on Mathematical and Computational Methods for Complex Social Systems, virtual

2019
- Minisymposium on Mean-field Models for Large Interacting Agent Systems
  SIAM Conference on Analysis of PDEs, La Quinta, CA
- Session on Mathematical Biology: Multi-Scale Modeling of Complex Biological Systems
  AMS Western Sectional Meeting, Riverside, CA
- Minisymposium on Dynamics of Democracy
  SIAM Conference on Applied Dynamical Systems, Snowbird, UT

Invited Talks and Participation (continued)

- Session on Advances in Mathematical Biology
  Association for Women in Mathematics Research Symposium, Houston, TX
- Session on Agent-Based Modeling in Biological and Social Systems
  Joint Mathematics Meeting, Baltimore, MD

2018
- Minisymposium on Modeling Cell Motility and Cytoskeleton Interactions
  SIAM Conference on Life Sciences, Minneapolis, Minnesota
- Session on Modeling Biological Rhythms
  MAA MathFest, Denver, CO
- Workshop Celebrating Diversity: Nonlinear Dynamics and Complex Systems
  SIAM Annual Meeting, Portland, OR
- Minisymposium on Spatial Dynamics: Local and Global Results
  SIAM Conference on Nonlinear Waves and Coherent Strutures
- Session on Modeling Disease and Biological Processes
  Joint Mathematics Meeting, San Diego, CA

2017
- Society for Mathematical Biology, Salt Lake City, UT

2015
- Stochastic Neural Dynamics Workshop
  Organization for Computational Neuroscience, Prague, Czech Republic

Workshops

2021
- Workshop on Mathematics and Racial Justice
  Mathematical Sciences Research Institute, virtual
- Mathematical and Computational Approaches to Social Justice
  ICERM workshop (virtual)
- SMB Workshop on Equity, Diversity, and Inclusion
  Society for Mathematical Biology, virtual

2019
- Applied Mathematical Modeling with Topological Techniques
  ICERM workshop, Providence, RI
- Learning in Networks
  NIMBioS/DySoC Working Group, University of Tennessee, Knoxville, TN

2018
- Cultural Analytics
  IPAM workshop, Lake Arrowhead Conference Center, CA
- Agent-Based Modeling in Biological and Social Systems
  AMS Mathematics Research Community, West Greenwich, RI

2017
- Women Advancing Mathematical Biology Workshop
  MBI at The Ohio State University, Columbus, OH

Professional Activities

Service
- President, Southern California SIAM Regional Section
  Data Science Curricular Coherence Committee, Claremont Colleges
  Advisory board member, SIAM activity group on Quantitative Justice
  Academic Affairs Committee, HMC
  Faculty advisor, SIAM student chapter at HMC
  Red Sock Poster Judge, SIAM Conference on Applied Dynamical Systems

Professional Activities (continued)

Peer-review	Chaos Physica D SIAM Journal on Applied Dynamical Systems SIAM Review
Memberships	Society for Industrial and Applied Mathematics American Mathematical Society Mathematical Association of America Society for Mathematical Biology